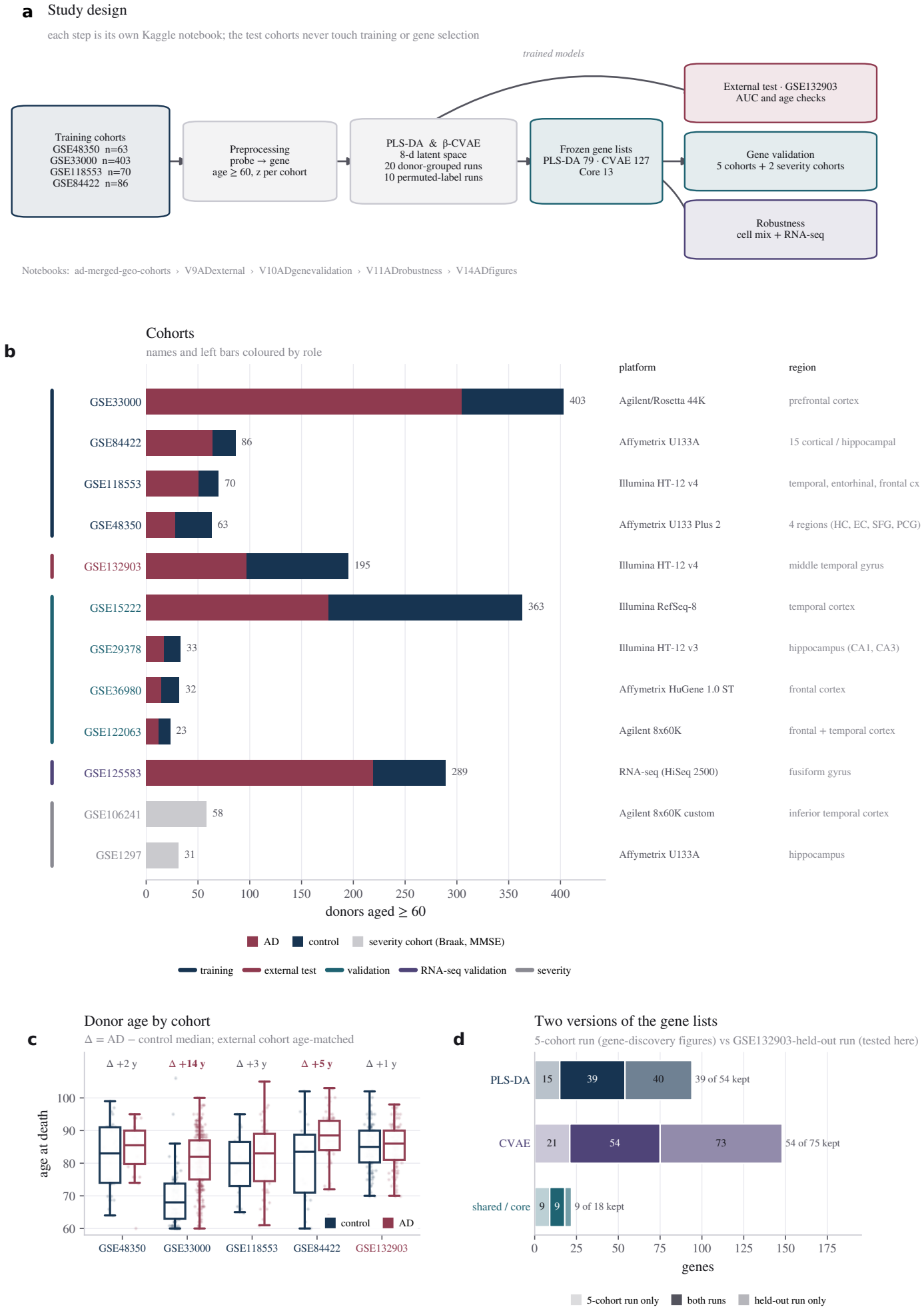
a
Study design
each step is its own Kaggle notebook; the test cohorts never touch training or gene selection
trained models
Training cohorts
GSE48350 n=63
GSE33000 n=403
GSE118553 n=70
GSE84422 n=86
Preprocessing
probe → gene
age ≥ 60, z per cohort
PLS-DA & β-CVAE
8-d latent space
20 donor-grouped runs
10 permuted-label runs
Frozen gene lists
PLS-DA 79 · CVAE 127
Core 13
External test · GSE132903
AUC and age checks
Gene validation
5 cohorts + 2 severity cohorts
Robustness
cell mix + RNA-seq
Notebooks: ad-merged-geo-cohorts › V9ADexternal › V10ADgenevalidation › V11ADrobustness › V14ADfigures
b
Cohorts
names and left bars coloured by role
platform
region
GSE33000
403
Agilent/Rosetta 44K
prefrontal cortex
GSE84422
86
Affymetrix U133A
15 cortical / hippocampal
GSE118553
70
Illumina HT-12 v4
temporal, entorhinal, frontal cx
GSE48350
63
Affymetrix U133 Plus 2
4 regions (HC, EC, SFG, PCG)
GSE132903
195
Illumina HT-12 v4
middle temporal gyrus
GSE15222
363
Illumina RefSeq-8
temporal cortex
GSE29378
33
Illumina HT-12 v3
hippocampus (CA1, CA3)
GSE36980
32
Affymetrix HuGene 1.0 ST
frontal cortex
GSE122063
23
Agilent 8x60K
frontal + temporal cortex
GSE125583
289
RNA-seq (HiSeq 2500)
fusiform gyrus
GSE106241
58
Agilent 8x60K custom
inferior temporal cortex
GSE1297
31
Affymetrix U133A
hippocampus
0
50
100
150
200
250
300
350
400
donors aged ≥ 60
AD
control
severity cohort (Braak, MMSE)
training
external test
validation
RNA-seq validation
severity
c
Donor age by cohort
Δ = AD − control median; external cohort age-matched
Δ +2 y
Δ +14 y
Δ +3 y
Δ +5 y
Δ +1 y
age at death
100
90
80
70
60
control
AD
GSE48350
GSE33000
GSE118553
GSE84422
GSE132903
d
Two versions of the gene lists
5-cohort run (gene-discovery figures) vs GSE132903-held-out run (tested here)
PLS-DA
15
39
40
39 of 54 kept
CVAE
21
54
73
54 of 75 kept
shared / core
9
9
9 of 18 kept
0
25
50
75
100
125
150
175
genes
5-cohort run only
both runs
held-out run only